

1 Problem Statement

Consider a neural network trained using the square loss function. Suppose the input to the neural network x is fixed. Assume that the learning rate for gradient descent α is also fixed.

Square loss is a function of the network output y and some target a . Consider two different targets a_1 and a_2 . Suppose gradient descent is applied $n\%$ of the time using target a_1 and $m\%$ of the time using target a_2 . As training continues, why would the output of the network when given input x be the expectation $n \cdot a_1 + m \cdot a_2$?